# Response to the reviewers (NEUCOM-D-26-16868)

Certified Correction of Object Hallucinations in Vision–Language Models: Guarantees, Gains and Failure Modes under a Pre-specified Protocol

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This document answers the comments of both reviewers of the earlier submission. All numbers are produced by the scripts and result files in the public repository (directory `conformal_followup/revision`); every table in the revised manuscript is generated from those files. Section and table numbers refer to the revised manuscript.

## Reviewer 1

| # | Reviewer point | What was done | Where |
|---|---|---|---|
| 1 | Proved and evaluated procedures differ (rank-indexed vs value-indexed) | Every reported number now uses a fixed **value** grid and a fixed repair gate, both read off the fit fold (score grid from out-of-fold fit-fold scores). The rank-indexed procedure is kept only as the conventional baseline (ladder rung L0) and in the known-population audit. | Method 4.2; Results 6.2, 6.6; `ccrc_rev.py` |
| 2 | Exchangeability does not justify the binomial step | Assumption 1 is now i.i.d. sampling of calibration units, independent of fit data; the proof is rewritten (emission pattern conditioning, i.i.d. draws from P(.\|emit)); a remark states that exchangeability alone gives a hypergeometric count. The empty-emission convention is stated with the definition of risk. | Setup 3.1; Thm 1 |
| 3 | Validity audit used data that participated in fitting/calibration | The "full item set" column is removed. The test fold is used for nothing else (no score, gate, start or method is selected on it); two test-fold statistics are reported, plus an exact audit in a known-population simulation (risk of the selected policy computed on 400k reference items). | Protocol 5; Results 6.6; Table audit |
| 4 | Gate q=0.10 chosen with knowledge of the benchmarks | Main analyses use (a) a gate fixed on a different benchmark at the target's calibration size (a full-size variant is also reported, and the size-matched rule was defined after seeing it fail on small cells), (b) a gate chosen on the fit fold, (c) a union bound over three gates (Corollaries 1-2). q=0.10 is kept only as an uncertified reference arm and is labelled so. | Method 4.2-4.3; Results 6.1 |
| 5 | Abort is partly created by the start of the sequence (k=33 vs 38) | k_start(e0) derived analytically (22, 38, 52, 65 at alpha=delta=0.10; Lemma 2) and the whole range e0 in {0..3}, c in {1.0,1.5} is ablated at alpha in {0.05,0.10,0.15}. Default e0=1. The AMBER "loss" shrinks from -10.2 to -0.3 points once the start is analytic, as the reviewer anticipated. | Results 6.2, 6.4; Tables start |

| # | Reviewer point | What was done | Where |
|---|---|---|---|
| 6 | Unit of splitting; image-grouped splits | Image identifiers of every cached item were recovered from benchmark metadata and verified against cached gold/object/category (all agree). POPE has six questions per image (75/100/250 images in the three cell sizes). All folds are unions of whole images. Intra-image error correlation is measured (about -0.04 to -0.09: slightly negative) and a known-population simulation shows the effect of positive correlation and the one-item-per-image fix. | Protocol 5; Method 4.4; Table cohort |
| 7 | "Independent evidence channel" overstated | Replaced by "distinct evidence source" throughout; independence is stated to be neither assumed nor needed for validity, and distinctness is framed as an empirical condition for useful repair regions. The instantiation's dependence (detector score feeds channel 1) is stated. | Setup 3.3; Abstract; Discussion |
| 8 | Repair accounting: changed vs confirmed answers; polarity | New accounting: repaired mass, answers changed, wrong->right, right->wrong, agree-right, agree-wrong, and false-positive / false-negative errors corrected, for pooled and polarity-specific gates. 41-96% of repaired items are confirmations; the answer changes for 0.04-1.7% of test items; no false-positive error is corrected on POPE. The benefit is described as re-emission with detector confirmation plus a small correction effect. | Results 6.3; Tables acct, sym |
| 9 | Cohort changes (444/450, 591/600) | Cohort-flow table (items, images, detector values, reason for each exclusion: unparsed POPE object phrase, or non-existence AMBER queries). The primary analysis keeps all items under a fixed missing-evidence policy (zeroed grounding features, missing flag, never repaired); the grounded-only analysis is in the appendix. | Protocol 5; Table cohort; App. grounded |
| 10 | Repeated-split statistics framed as dataset uncertainty | Split-based p-values are removed. Gains carry 95% intervals from an outer image-level bootstrap (300 resamples x 20 splits) and the fraction of resamples with a positive gain. The interval excludes zero in 7 of 12 (cell, alpha) combinations. | Protocol 5; Results 6.1; Fig. forest |
| 11 | Theorem 2 unit and signed interference | Rewritten as an exact pointwise identity with an expected count over repair opportunities, a signed interference term, and the single-repair special case. | Sequential 8 |
| 12 | Notation; empty emission; mu known | Emission indicator defined; risk of empty emission is 0 and aborts count as zero coverage; alpha is a choice, mu an estimate, and a feasibility check needs a confidence bound on mu. | Setup 3.1-3.2 |
| - | CEBC (ACL 2026) not discussed | Verified on the ACL Anthology (read in full); compared on action space, controlled quantity, guarantee, revised output, calibration, detector role. Novelty claim narrowed and scoped to a literature cutoff date. | Related work; Section 7 |
| - | BCEA reference outdated (v4) | Updated to v4 (21 Jul 2026); technical description and numbers corrected: certified Bonferroni-corrected CP over a fixed grid; 0.23 -> 0.32 certified at alpha=0.10; 0.28 -> 0.37 is the uncertified practical variant. The former composition experiment is removed rather than rerun. | Section 7 |
| - | Bibliography updates | Learn-then-Test (Annals of Applied Statistics 2025), MME (NeurIPS 2025 D&B), reference [22] (DOI added, malformed "In press" note removed), Kotte v2 and BCEA authors added; six further related works verified. Absolute claims ("nothing in the literature") scoped to the surveyed literature and a date. | refs.bib |

## What changed in the conclusions

- Signals were extracted for 1,500 more POPE items (250 new images; POPE-3000 is the whole adversarial split) and 2,500 more AMBER items (AMBER-3000, 957 images) on a Colab T4, so there are six cells and twelve (cell, alpha) combinations.

- With the gate fixed on external data at the target's calibration size, the gain over filtering is +0.8 to +7.3 points, positive in all twelve, with an image-bootstrap interval excluding zero in seven. It is about one point with 150-200 calibration items and up to +7.3 with 500 or more.
- The fit-fold gate and the union bound are valid; they help with 500+ calibration items and cost coverage with 150-200.
- Repair accounting depends on the gate: with a small gate under 0.5% of test items change; with a large gate 2.4-5.7% change, 81-93% of the changes are right, and 4-5% of false positives are corrected.
- The AMBER loss reported before was produced by the start of the sequence (CCRC aborted on 72% of splits under the conventional protocol).

## Reviewer 2

| # | Reviewer point | Response | Where |
|---|---|---|---|
| W1 | Novelty: a combination of existing parts; Remark 2 shows the guarantee is indifferent to the answer's source | The introduction now says plainly that Theorem 1 is an application of fixed-sequence testing and that no new statistical technique is claimed. The contribution is stated as the formulation (a risk certificate over a mixture of the model's and a second predictor's answers, which escapes the abstention floor), a protocol in which the proved and evaluated procedures coincide, and measurements of what the mechanism contributes. | Intro, "What is and is not new" |
| W2 | Theorem proved for value-indexed families but experiments use rank-indexed ones | Every reported number now uses a value-indexed family (thresholds and gate from the fit fold). The rank-indexed variant is kept only as the conventional baseline and in a known-population audit (it is also conservative there). | Method 4.2-4.5; Results 6.2, 6.8 |
| W3 | Channel 1 includes the detector score; test a detector-free channel 1 or a second detector | New analysis with every detector feature removed from channel 1 (detector used only for repair). CCRC gain over the matching filter is positive in all 12 combinations (+0.3 to +46); absolute coverage is below the full pipeline in 10 of 12 combinations (by 0.1-35 points). A second detector was not run; stated as a limitation. | Results 6.9, Table detfree; Discussion |
| W4 | Scope narrower than the title | Title changed to "... Object Hallucinations in Vision-Language Models"; abstract and introduction state the object-existence scope; Discussion adds which evidence sources could extend the method to attributes, counts and relations, and what happens without one (reduces to filtering). | Title; Abstract; Discussion |
| W5 | Remedies for the AMBER failure left untested | Both implemented. The analytic start (e0 >= 1) is ablated over e0 in {0,..,3} and c in {1.0, 1.5} at three risk targets, and removes the AMBER loss (the conventional start made CCRC abort on 72% of splits). A delta-spending search over three starts at delta/3 is also tested; it is below the single analytic start in all 12 combinations (by 0.7-13 points). | Results 6.2, 6.4; App. multi-start |
| W6 | Framing of the sequential result | Section and abstract reworded: "a matched-prefix test", "a small experiment suggests"; the under-power (n=121, MDE above the observed effect, Hodges-Lehmann 0) is stated next to the result and in the introduction. | Sequential 8; Abstract; Intro |
| W7 | Intervals are over splits; evaluate on held-out data | Outer image-level bootstrap intervals replace split-based intervals; two benchmarks were extended (POPE to its full 500 images, AMBER to 957 images). A new held-out-benchmark analysis fits the score, grid and gate on another benchmark and uses the target only for calibration and test; the certificate holds (confident violations <= 0.03 in nine of ten combinations; the exceptions are the smallest cells, where 75-98% of splits abort). | Protocol 5; Results 6.1, 6.9, Table transfer |

| # | Reviewer point | Response | Where |
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| W8 | Missing related work: learning to defer and certified cascades | New paragraph positioning CCRC against learning to defer (Madras et al.; Mozannar and Sontag), rejection learning, FrugalGPT-style cascades and recent guaranteed cascades (Trust or Escalate and four preprints), with the structural difference (second predictor = different-modality detector whose margin gates substitution; no learned policy). All entries verified on primary sources. | Related work |
| W9 | Length (about 70 pages, 23,500 words); abstract above 250 words | Main text is now about 26 pages (11 pt, 1.5 spacing, tables and figures included); abstract is 230 words; secondary analyses (gain decomposition, multi-start search, detector-free benchmarks, all-arm audit, ladder at other targets, grounded-only cohort) are in the appendix; headline numbers are restated less. | Whole paper |
| W10 | Academic writing: abstraction, anecdotal opening, failures dominate | The introduction opens with the problem statement and refers to the schematic (Figure 1), states the main result early, and presents the failure modes as boundaries; the revision-history framing is removed. | Introduction |
| W11 | Contrastive constructions and rhetorical asides | A pass removed most "A rather than B"/"A, not B" constructions and asides; ten remaining technical uses are kept where the contrast is the content (for example hypergeometric versus binomial). | Throughout |
| W12 | Internal inconsistencies (four of five settings, 8 of 10 cells, 9.0% vs 8.1-point floor, 44% coverage) | Every number in the text was checked against the generated tables; abstract, introduction and results use one set of cells (six cells, twelve combinations). The unsupported "competitive with published results" claim is removed. | Whole paper |
| O1 | Too many contributions/findings | Four design decisions (the former six contributions), six findings, each tied to one section. | Introduction |
| O2 | Doubled periods in run-in headings | Removed (the Elsevier class appends its own period). | Source |
| O3 | Highlights missing | Added (five bullets). | manuscript (highlights file) |